LESSON 13 · SESSION 1 · EXPLORE EQUIVALENT RATIOS

Equal Groups Make Equivalent Ratios

E L A + M A T H · W O R D S T U D Y

Mr. H Math · Chandler Park Academy · Day 21

C C S S
6.RP.A.3a
L.6.4
SMP 2,7,8

WRITE YOUR NAME — one letter per box, straight and neat

Date: _____ **Score: ___ / 10**

On Monday you said 6 tacos for every 3 guests is the same deal as 2 tacos for every 1 guest. Today you prove it. Veda makes 3 batches of henna paste for Diwali — one batch, then another identical batch, then another. Equal groups is the whole idea: what you do to one quantity you do to the other, every single time.

DO NOW · SILENT START — 3 min

Which one doesn't belong? **2 : 3** · **4 : 6** · **6 : 9** · **3 : 2** Circle it and write why in one sentence.

TURN & TALK — 2 min

Share your circle. If you and your partner circled the same card for different reasons, write down both reasons — one of them is about order, one is about equal groups.

Frames: "_____ doesn't belong because _____." · "_____ and _____ are equivalent because _____."

Section A · Guess the Word

Fill in the blanks to find the word for ratios that express the same comparison.

	Q	U	I	V	A	L		N	T

The SAME missing letter goes in both dashed boxes. Write it here: _____

3 : 2 and 9 : 6 are _____ ratios.

You build one by **combining equal groups** — which is the same as multiplying **both** quantities by the same number. Adding the same number to both does *not* work.

Section B · Meet the Equal-Groups Family

Use the word bank to complete each sentence.

equivalent · **equal groups** · **batch** · **multiply**

1. Ratios that express the same comparison are _____ ratios.
 Example: 3 : 2, 6 : 4, and 9 : 6.
2. One complete copy of a recipe is a _____.
 Example: 3 batches = 3 identical copies, not 3 more of each.
3. Combining _____ of the same size builds an equivalent ratio.
 Example: one group of 3 and 2, then another, then another.
4. To jump straight to an equivalent ratio, _____ both quantities by the same number.
 Example: 3 : 2, both × 3, gives 9 : 6.

Section C · Try It: Three Batches of Henna Paste

One batch uses 3 tablespoons of henna powder for every 2 teaspoons of water. Study the worked example, then build your own equal groups.

WORKED EXAMPLE

1 batch = 3 powder : 2 water

●●●□□ ●●●□□ ●●●□□

3 batches → 3×3 = 9 powder, 3×2 = 6 water

3 : 2 = 9 : 6

↑ both quantities × 3 — not + 3

YOUR TURN

Start with **3 circles and 2 squares**. Draw **4 equal groups**, then finish the lines.

4 groups → _____ circles : _____ squares

Is **5 : 4** equivalent to **10 : 8**? _____ Why?

REFINE · SUMMARY — Equivalent ratios are the same comparison wearing different numbers. Build one by adding an identical group, or jump straight there by multiplying **both** quantities by the same number — the one move that never works is adding the same number to both. Next: two tools that hold a whole family of equivalent ratios at once — the **double number line** and the **ratio table**.

☆ Equal groups, every time! ☆

E L A + M A T H